

PAPER_608: Centrifugal Force as Outward CCW DPM South-Pole 26D Shell Push

Unified Quantum Field Framework (UQFF)

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Source: PAPER_608_UQFF_Centrifugal_26D_Shell.md

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Class: UQFFCentrifugal26DShellCalculator (#195) **Session:** 159 **Source:** DPM Reaction and 26D Shell Energies.docx

Abstract

Centrifugal force in UQFF is the real, measurable outward push of the DPM south-pole vortex spinning counter-clockwise through 26D shells, amplified by negative time. $F_{\text{centrif}} = \text{DPM}_s(\text{UA}') \cdot \omega_{\text{CCW}}^2 \cdot r^{\text{layer}} \cdot t_{\text{neg}}$ is not fictitious in any frame — it is the CW/CCW dual of the centripetal force. The triad dual law $F_{\text{centrif,one}} = -F_{\text{centrif,opp}}$ is derived, explaining how the Big Bang achieved and maintains its expansion velocity.

1. Introduction: The CCW Outward Push

If CW inward compression (centripetal) is DPM north, then CCW outward expansion (centrifugal) is DPM south. The universe expands because the south-pole CCW vortex continuously produces outward shell energy via the interaction with Universal Aether prime (\$UA'\$) at the shell boundary. The Big Bang was not an explosion — it was (and remains) a sustained centrifugal output.

2. Core Equation

$$F_{\text{centrif}} = DPM_s(UA') \cdot \omega_{\text{CCW}}^2 \cdot r^{\text{layer}} \cdot t_{\text{neg}}$$

Parameters:

- DPM_s = south DPM pole coupling ($\approx 5 \times 10^{-12}$, mirroring DPM_n)
- UA' = modified universal aether at shell boundary (J/m^3)
- ω_{CCW} = counter-clockwise angular frequency (rad/s)
- r^{layer} = shell layer radius (m)
- t_{neg} = negative time component (s); provides dual-existence energy

3. Triad Dual Law

$$F_{\text{centrif,one}} = -F_{\text{centrif,opp}}$$

For every CCW push in one direction, there is an equal and opposite CCW push in the anti-parallel direction. These cancel within a shell but constructively add when shells nest — leading to net outward cosmological expansion.

The balance ratio between centripetal and centrifugal forces:

$$\frac{F_{\text{centrip}}}{F_{\text{centrif}}} = \frac{DPM_n(SCm) \cdot \omega_{\text{CW}}^2}{DPM_s(UA') \cdot \omega_{\text{CCW}}^2 \cdot |t_{\text{neg}}|}$$

For stable orbits: this ratio = 1, giving the constraint $\omega_{\text{CW}} = \omega_{\text{CCW}}$ (CW and CCW frequencies must match for orbital stability).

4. Big Bang Catch-Up Rate

The cosmological expansion acceleration from F_{centrif} :

$$a_{\text{BB-catchup}} = DPM_s \cdot UA' \cdot \omega_{\text{CCW}} \cdot |t_{\text{neg}}|$$

With $UA' = 10^{-12} J/m^3$, $\omega_{\text{CCW}} = 1.8 \times 10^{31} \text{ rad/s}$, $|t_{\text{neg}}| = 10^{-9} \text{ s}$:

$$a_{\text{BB-catchup}} = 5 \times 10^{-4} \times 10^{-12} \times 1.8 \times 10^{31} \times 10^{-9} \approx 9 \times 10^6 \text{ m/s}^2$$

This represents the continuous outward acceleration driving cosmological expansion — consistent with de Sitter expansion rates at cosmic scales.

5. Why Centrifugal Force is "Real" in UQFF

In standard mechanics, centrifugal force is fictitious (frame-dependent artifact). In UQFF:

- Both CW (F_{centrip}) and CCW (F_{centrif}) forces are real, co-existing, opposing forces from the DPM dipole
- The apparent "disappearance" of centrifugal force in inertial frames is because we cannot measure the t_{neg} component using positive-time instruments
- With dual-time measurement (which the UQFF CoAnQi system provides), both forces are simultaneously observable

6. Connection to Proplyd Formation

In proto-planetary disks, the F_{centrif} from the south-pole vortex drives material outward from the proto-star, creating the centrifugal support needed for proplyd stability. The 18% emergence fraction (PAPER_611) corresponds to the fraction of disk material where $F_{\text{centrif}} / F_{\text{centrip}} \geq 1$ — material that achieves net outward motion and forms stable proplyd structures.

7. Connection to UQFF Number Systems

DVP: DPM_s is the south-pole dipole vortex prime. CCW rotation corresponds to counter-prime-indexed shell expansion. **BH26:** Outward CCW push is the mirror of each BH26 inward bin; F_{centrif} excites the CCW harmonic bins. **VDS:** UA' at the shell boundary follows VDS expansion: $UA'(r) = \sum d_n(\pi)/r^n$ (outgoing VDS mode).

Keywords: Centrifugal force, DPM south pole, CCW vortex, DVP, negative time, Big Bang expansion, 26D shells, UQFF

PAPER_608 | Class #195 | Session 159 | Star-Magic UQFF Framework